NSIN

National Security Innovation Network

Organizational Overview

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National Security Innovation Network (NSIN)
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Fast Facts

- A small Department of Defense office comprised of civilians, military, and contractor personnel;
- Formed in October 2016 as the MD5 National Security Technology Accelerator;
- Currently report through the Defense Innovation Unit (DIU) to the Under Secretary of Defense for Research and Engineering;
- In partnership with 38 top-tier universities and five commercial incubators and accelerators
- Operating in 35 States as of February 2020

Our Mission:

WE BUILD NETWORKS OF INNOVATORS TO GENERATE NEW SOLUTIONS FOR NATIONAL SECURITY PROBLEMS

The Theory Underpinning Our Work:

- DoD's current model for problem-solving is expensive, slow, and inefficient;
- The new complexity of problems the DoD faces, and the speed at which solutions are required, means that new problem-solvers must be introduced into the system;
- Adding the intellectual diversity from non-traditional solution providers (i.e., the academic and early-stage venture communities) is a necessary step to solving DoD problems cheaper, faster, and better;
- A network approach enables persistent problem-solving capability for the Department of Defense that helps bridge the civil-military divide and improve outcomes for the Department of Defense

Our Customers: We Help Solve Their Problems

- Major Commands within the Army, Air Force, Navy, and Marine Corps
- Combatant Commands
- Tactical Units in each Military Service
- Joint Staff
- Office of the Secretary of Defense
- DoD Field Agencies and Defense Activities
- DoD Labs

We're Building A Network of National Security Problem-Solving Communities

Region	Hub City	NSIN Regional Director
Region I (Northeast)	Boston, MA	**Mr. Matt Merighi** mmerighi@nsin.us
Region II (Mid-Atlantic)	New York City	**Mr. Dan Madden** dmadden@nsin.us
Region III (National Capital)	Washington, D.C.	**Ms. Jennifer Bird** jbird@nsin.us
Region IV (Southeast)	Raleigh, NC	**Dr. Tommy Sowers** tsowers@nsin.us
Region V (Great Lakes)	Chicago, IL	Coming in September 2020
Region VI (Southwest)	Austin, TX	**Mr. Jim Rabuck** jrabuck@nsin.us
Region VII (Midwest)	St. Louis, MO	Coming in Winter 2021
Region VIII (Rocky Mountain)	Denver, CO	Coming in Fall 2020
Region IX (Pacific-North)	San Francisco, CA	Coming in Fall 2020
Region IX (Pacific-South)	San Diego, CA	**Mr. Jesse Gipe** jgipe@nsin.us
Region X (Northwest)	Seattle, WA	Coming in Spring 2020

University Program Director @ University of Nebraska

Wade B. Watts

Responsibility

As University Program Director, Mr. Watts helps build and grow a network of national security problem solvers at the University of Nebraska by serving as the principal liaison between the University and the DoD for all matters pertaining to NSIN.

Academics

Degrees and Certifications

- Master of Public Administration
- Master of Airpower Art & Science
- B.S. Computer Science
- Project Management Professional (PMP)

Graduate-Level Adjunct Faculty

- Creighton University
  - Business Consulting
  - Project Management
- Midland University
  - Capstone Mentor

Experience

Former Air Force Officer

- Electronic Warfare Officer
- Extensive experience in C6ISR, Electronic Combat, Combat Planning
- Served at USSTRATCOM and 55th Wing

Commercial Experience

- Program Manager (Systems Engineering Contract) for USSTRATCOM Global Innovation and Strategy Center (GISC); Oversight of Intern Program with UNO/PKI
- Management Consulting
- Program Management Office Director
- Companies: Booz Allen Hamilton, Ameritas, Fiserv (formerly First Data), Omaha Chamber of Commerce

Entrepreneur (Previous)

- Management Consulting CEO
- International Business Brokerage COO
- Real Estate Partner
- Photography Services

Contextualizing Our Innovation Model Within the Defense Innovation Ecosystem

Game-Changing, Basic Research (Slow, Big Bets):

Strategy-Driven, Solution Development (Fast, Big Bets):

End User-Driven,
Rapid Solution Development (Fast, Little Bets):

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Requirements-Driven,
Rapid Solution Discovery (Buy vs. Build):

What We Do to Accomplish Our Mission: Our Work

We…

Create New Opportunities for National Security Service by:

- Building models of service that account for generational and cultural differences between the military, academic, and venture communities; and
- Providing flexible pathways to official service within the Department of Defense.

Solve National Security Problems by Collaborating with Partners from the Academic and Venture Communities by:

- Engaging new problem-solvers in collision events with DoD customers that generate novel concepts and solutions; and
- Building a national network of problem-solving ecosystems that leverage the competitive advantages of regions and commercial innovation hubs for DoD customers.

Accelerate the Adoption of Novel Concepts and Solutions by:

- Facilitating engagement with DoD end users and transition partners to stimulate dual-use venture growth; and
- Improving Technology Transfer and Transition (T3) rates for DoD Lab technology through dual-use commercialization via early-stage ventures.

How NSIN Partners with Universities

We…

- Provide direct access to DoD problems and agencies to facilitate collaborative problem-solving (i.e., concept development) with faculty and students;
- Create bespoke programming opportunities through direct funding that leverage a partner university's competitive advantages or competencies relevant to the DoD;
- Identify dual-use technology that a university partner has developed which may be relevant to DoD end user needs;
- Execute core NSIN programming, such as Hacking for Defense or Hackathons, that expose students to DoD problem sets in an open source, unclassified environment;
- Recruit students, entrepreneurs, and technologists for applied, project-based internships, fellowships, and contract work opportunities.

Core Programs and Services

National Service

- **X-Force Fellowship and X-Force Volunteers** Embeds recent graduates and current students with a tactical unit to solve DoD end user problems ranging in complexity from follow-on H4D prototyping to mobile app development
- **X-Force Capstone** Run like X-Force volunteers, but in conjunction with a CAPSTONE or other course.
- **Hirethon**
  Facilitates access for DoD organizations with direct or expedited hiring authority to talents of STEM, cyber, and venture talent through third-part events (e.g., HackGT) or extant NSIN programs (e.g., H4D Closing Pitch Event)
- **Defense Civilian Auxiliary Corps**
  Identifies personnel with STEM, cyber, and entrepreneurial backgrounds for future, term service with the DoD by placing them through a dedicated academic course of study that results in a graduate-level certificate or degree. Currently piloting the curriculum development with MIT and UC-Berkeley
- **Technology and National Security Fellowship Pilot**
  Authorized by the FY20 NDAA, the Fellowship permits placement of exceptionally qualified STEM graduates in the offices of Members of Congress and presidentially appointed, senate confirmed officials within the DoD. The inaugural pilot will be executed in FY20 with six individuals. Positions will be compensated at the GS-10 pay band.

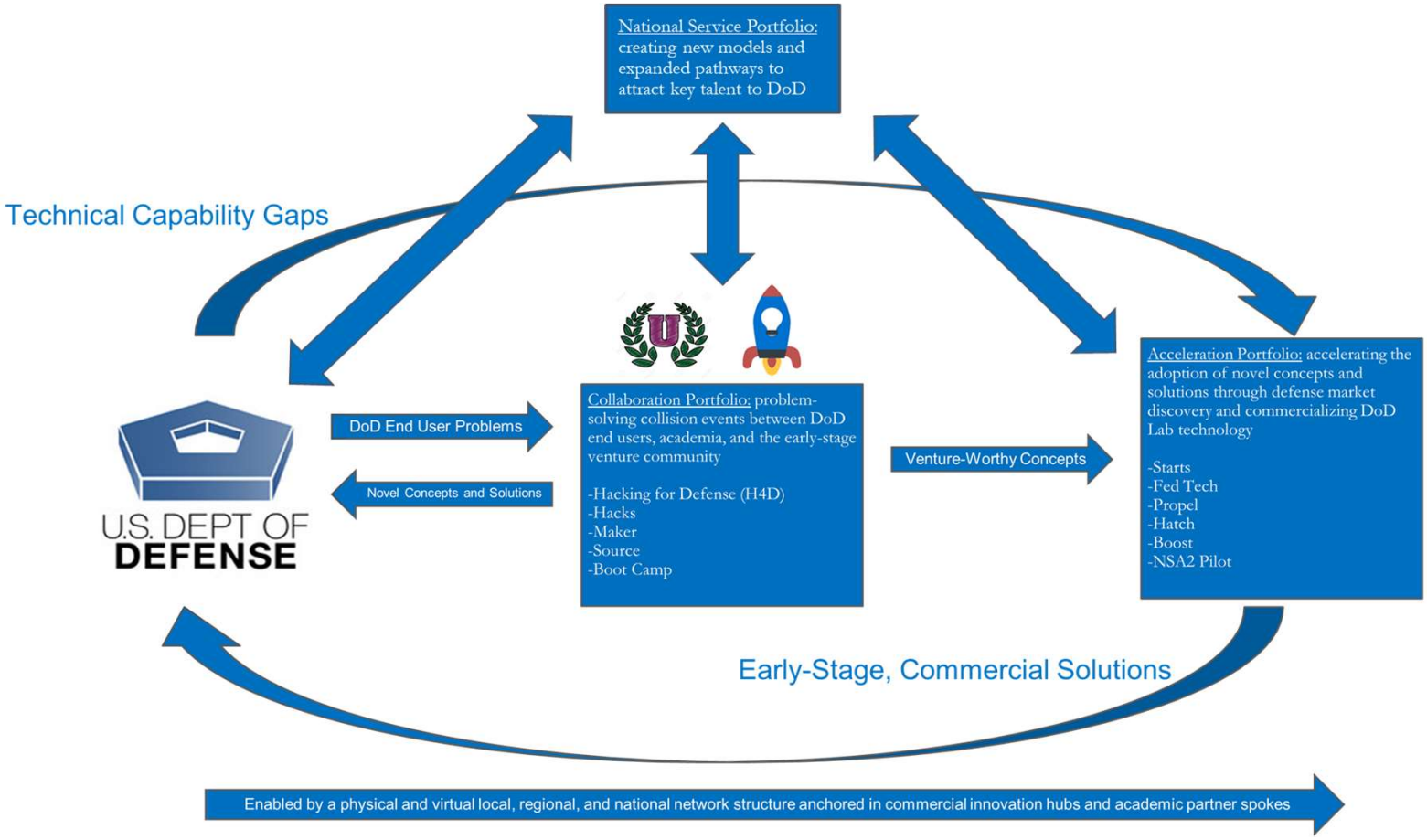
Collaboration

- **Defense Innovation Network**
  Web-based, defense entrepreneur community that incorporates people, problems, tech, resources, and solution concepts in a single space
- **Hacks**
  54-hr hackathons bringing together civilian and military innovators to build prototypes that address national security challenges
- **Hacking for Defense (H4D)**
  University-based venture-creation course focusing on national security problems from DoD sponsors
- **Source**
  Provides opportunities for Service Members to provide bottom-up, direct feedback to senior leaders for utilization and adoption through virtual innovation challenges
- **Proof of Concept Centers (Maker)**
  Prototyping and experimentation utilizing university partner resources for translating solution concepts developed by defense innovators to applications suitable for transition or commercialization
- **Boot Camp**
  5-day course that teaches Service Members how to develop MVP solution concepts in a tactical environment for problems that directly affect their command

Acceleration

- **Starts**
  Program that provides unique opportunities to early-stage venture teams for access to public and private capital and facilitate follow-on funding by DoD problem sponsors
- **Defense Innovation Accelerator (DIA)**
  DoD Lab-to-market tech commercialization program that facilitates the development of startups in the national interest
- **Boost**
  Sponsors SBIR Phase I Special Topics in conjunction with transition partner commands to develop early-stage prototypes and proofs of concepts for DoD capability gaps
- **National Security Academic Accelerator (NSA2)**
  A University based innovation program that seeks to increase the number of dual use university startups aimed at developing solutions for national security challenges
- **Propel**
  Partners with commercial startup accelerators to lower barriers to entry for dual use ventures and lowers risk for DoD end users in discovering novel solutions

Operationalizing NSIN's Approach



What Does Success Look Like?

- *NSIN Hacks San Diego:* the USD(I) Defense Intelligence Innovation Office (DI2O) and U.S. Army Special Operations Command served as DoD partners for an NSIN Hackathon in San Diego which occurred in 3Q19. Since then, four of five winning teams have formed companies, raised more than $2.3M in capital (including an SBIR award), and two of them are executing prototype OTAs through USASOC.

- *Pison and NSIN Propel:* Pison is a Boston-based hardware start-up with a solution for human-machine interface. Their wearable device detects the movement of the user's hand to enable gesture control of drones and software (their two initial applications). Prior to their involvement with NSIN, they were pursuing the commercial wearables market or a medical use case. Over the past eight months, they have built relationships with USSOCOM and USD(I) and won several SBIRs with a total value of more than $2 million. They are currently working towards securing an Air Force Strategic SBIR, with a potential value in excess of $20 million.

- *Direct Kinetic Solutions and NSIN Defense Innovation Accelerator (DIA)*: degraded, miniature isotope that provides battery life for 30-100 years. Technology originated from ARL-Adelphi; company formed in January 2019, already raised more than $800K, and is in the final stages of executing a licensing agreement w/ ARL. Currently prototype is at TRL-4.

NSIN & UC SAN DIEGO PRESENT
OFF THE GRID HACKATHON
Tactical computing capabilities in austere environments

NSIN

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